System Development of a Low-Cost Wireless Open-Source Force Myography Band for Hand Gesture Data Collection

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***Abstract*—This paper presents the design and development of a low-cost wireless force myography band for hand gesture data collection and offline classification. The system uses force-sensitive resistors, embedded signal acquisition, and wireless data transmission to record radial muscle forces in the forearm that occur during hand gestures. Collected data is then processed offline to extract features and evaluate gesture classification performance. This paper will discuss the motivation for developing such a system, outline key hardware and software design decisions, and describe preliminary validation that was done.**

I. Introduction

Control strategies for upper-limb prostheses have dramatically evolved in the last two decades and provide possibilities for increased intuitive control and patient functional outcomes [1], [2]. A non-invasive sensing mechanism widely used since the 1960s for muscle-powered prostheses is surface electromyography (sEMG), which uses electrodes to sense electrical activity generated by muscle action potentials under the skin [3]. However, a significant barrier to widespread consumer and research adoption of sEMG is cost. Widely used commercially available research systems such as the Delsys Trigno are priced in the range of tens of thousands of U.S. dollars, which limits accessibility outside research settings [4]. As a result, there exists a growing interest in exploring lower-cost sensing modalities in recent years. Force myography (FMG) is another emerging technique for monitoring muscle activity that measures changes in pressure exerted by muscles on the skin surface. FMG is considered a complementary sensing technique that offers similar performance in upper limb gesture decoding, with advantages including cost-effectiveness, high signal-to-noise ratio (SNR), and robustness to muscle fatigue, hair, and sweat [5], [6]. In creating a device that is accessible to the everyday consumer, FMG is advantageous in that it can utilize commercially available, off-the-shelf pressure sensors as a method for signal acquisition. The conditioning circuitry for these sensors do not require expensive or complex configurations, reducing the amount of components needed and keeping overall system costs low [7]. Leveraging these properties, we developed a low-cost, wireless, open-source FMG band for hand gesture data collection and classification. To support user mobility in research or environments beyond the laboratory, the system leverages the onboard Bluetooth connectivity of an ESP32-S3 microcontroller for wireless data transmission to a local computer during data collection, while also supporting USB cable-based data transfer as a reliable alternative.

Compared to prior systems that often require heavier external equipment, additional acquisition hardware, or more complex circuitry, this design emphasizes a lightweight and compact hardware setup suitable for portable force myography data collection.

II. Design Targets & Constraints

We adopted an iterative design approach to the FMG band development that prioritized the following technical targets:

- **Accessibility:** Maintain a low total build cost under $300.00 using primarily off-the-shelf components.
- **Low power operation:** Enable the system to be battery-powered within a 3–5 V supply voltage range [8].
- **Simple circuitry and data transfer:** Reduce circuit complexity and minimize the external hardware required for assembly and data collection. Previously developed systems commonly rely on costly data acquisition (DAQ) hardware, such as National Instruments' Data Acquisition System (NI DAQ) [8]. This design forgoes the need for an external DAQ by enabling data transfer via the onboard analog to digital converter (ADC) to either Bluetooth or USB for data collection.
- **Sensor modularity:** Support a modular sensor layout that allows sensors to be added, removed, or repositioned. This improves adaptability across different forearm sizes and provides flexibility for future expansion through interchangeable sensor modules, allowing additional sensing modalities to be integrated alongside FMG.

A prototype was considered successful if it met these design targets while remaining consistent with the required sampling rate for static gesture prediction identified in prior force myography literature [2]. Together, these constraints guided the development of a system that balances affordability, portability, power efficiency, and modularity.

III. Design

A. Electrical Design

The system consists of three primary electronic hardware subsystems: (1) a motherboard built around the ESP32-TinyS3 microcontroller, (2) sensor units (termed daughterboards) that house the force-sensitive resistor and ADC for each sensing channel, and (3) a LiPo battery. Each subsystem was either selected from off-the-shelf components or custom designed to meet the targets outlined in Section II. A high level overview of how subsystems interact is shown in Figure 1.

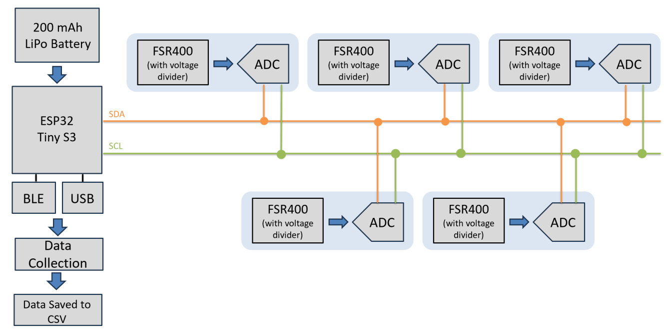

Figure 1. System-level diagram. This particular example has 5 FMG sensor modules connected via I^2C bus line to the TinyS3 microcontroller.

To satisfy the design targets for both USB and Bluetooth connectivity, the system uses Unexpected Maker's ESP32-TinyS3 microcontroller [9]. The TinyS3 supports 3.3V and 5V power inputs. 3.3V operation was chosen to reduce overall power consumption.

The TinyS3 also supports built-in FreeRTOS through the ESP32-S3 development framework. This enables the firmware to manage multiple software tasks, such as sensor acquisition and data transmission, while operating within the resource constraints of the embedded hardware.

For this prototype, a custom PCB (Figure 2 and 3) was designed for interfacing with the TinyS3 MCU soldered directly onto the board. This assembly is what we defined as the motherboard. This provided direct access to ESP32-S3 peripherals and external connections to other sensing daughterboards shown in Figure 4, 5, and 6. Daughterboard connections were managed by a 4-pin JST connector that communicated via I^2C, allowing additional units to be connected in series through matching JST interfaces. I^2C was selected because it requires minimal wiring for multiple sensor modules on a shared bus.

In the literature, there is no consistent standard that has been adopted for sensor quantity and sampling rate required for effective static hand gesture prediction [1]. Some studies that involve development of low-density FMG systems [1, 10, 11] have shown the usage of three to eight sensors. In our current configuration, the system can support up to 9 sensors while maintaining a sampling rate of 10 Hz. This rate falls within the 6 Hz to 1 kHz range reported in prior FMG studies [5]. Since the

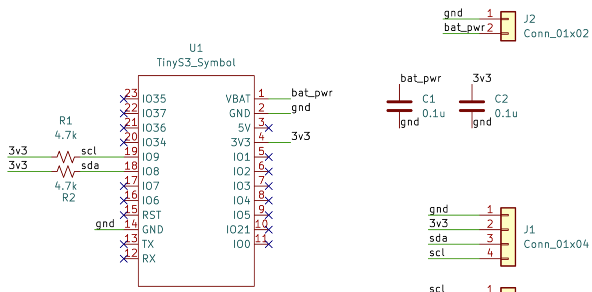

Figure 2. Schematic of motherboard.

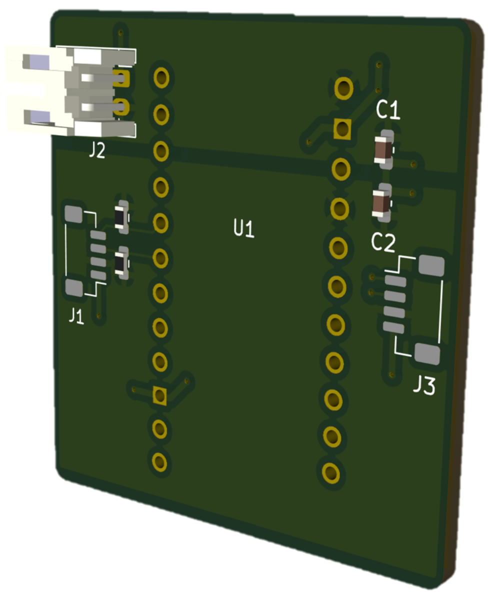

Figure 3. Microcontroller motherboard.

frequency of human hand motion is typically below 4.5 Hz [12], the selected sampling rate of 10 Hz is sufficient for capturing relevant gesture dynamics.

A 2-pin JST-PH connector was also included to support direct connection to a Lithium-ion battery. This allows the prototype to operate as a portable, battery-powered system rather than relying on a wired bench supply or USB power. Decoupling capacitors lie between the power and ground rails to improve voltage stability and reduce noise during operation.

FMG systems inherently rely upon force or pressure sensors to derive control signals from muscle activity [8]. For this project, we utilize Interlink Electronic's FSR400 sensors, which are polymer thick film (PTF) devices that alter their resistance based on the force exerted by a limb's active area on the sensor's surface. As the applied force increases, the sensor's resistance decreases [13]. These force-sensitive resistors are known for their reliability and sensitivity to changes in pressure. The force range of these sensors spans from 0.2–20 N [14].

Since FSRs are passive components, additional circuitry is needed to convert their resistance changes into a measurable analog signal. As viewed in Figure 4, a voltage divider (VD) circuit is used to create a potential difference between ground and the FSR output, allowing the sensor's resistance variations to be translated into a corresponding voltage [8]. The choice of the fixed resistor in the divider plays a critical role in shaping the sensor's response by limiting current flow and defining the voltage range. To accurately capture muscle contractions, which generate forces between 0–400 grams (0–4N), a 7.5 kΩ resistor was selected based on manufacturer guidelines [14]. This value optimizes the linearity of the output voltage within the target force range of 0–4N [8], [12].

Since the force-sensitive resistors produce analog voltage signals, an external analog-to-digital converter is required before the data can be processed by the microcontroller. The ADC121C021 by Texas Instruments, a 12-bit I^2C-interface ADC, was selected to digitize each sensor signal and transmit the resulting measurements to the MCU over the shared I^2C bus. The device is compatible with the system's targeted 3.3 V operating range, supporting low-power operation while avoiding the need for additional voltage-level shifting. Its compact size, simple interface, and support for multi-device I^2C communication also make it well suited for a modular sensor architecture.

Since the sensing daughterboard modules communicate with the MCU over a shared I^2C bus, each FMG sensor module must have a unique slave address to avoid bus conflicts. The ADC121C021 supports nine user-configurable addressing through

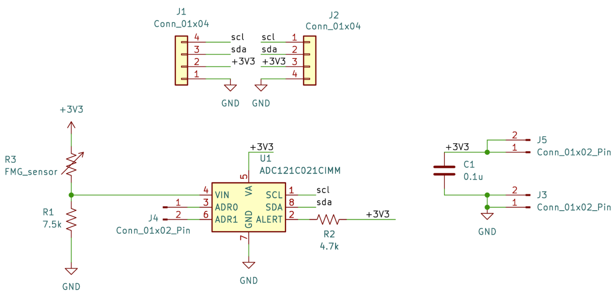

Figure 4. Schematic overview of a single daughterboard module. Each module consists of a VD outputting to an ADC121C021 I^2C-interface ADC converter, wired to the MCU's I^2C bus line. 3V3 battery supply and GND connected to decoupling capacitors for stability.

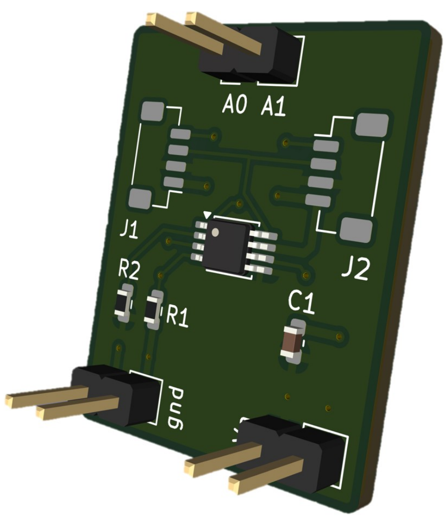

Figure 5. PCB front-view of a single force sensor module daughterboard. Includes the ADC121C021 component at the center. Jumper pins are routed to the address pins (A0 and A1) of the ADC, which could either float or be connected to GND or VA for slave address configuration. J1 and J2 label where the 4-pin JST connectors would be placed.

its A0 and A1 address pins [15]. By tying these pins using female-female jumper wires to different logic states, such as VA, GND, or leaving them floating, each ADC can be assigned a distinct I^2C 7-bit identifier address. This allows multiple sensor modules to be connected on the same bus while still enabling the MCU to identify and read from each sensor individually.

Figure 6. PCB exploded back-view of a single force sensor module daughterboard, consisting of an FSR400 sensor and bumper.

B. Software Design

To support our design target of sensor modularity, specifically in the case where there is a higher quantity of sensors or a potential change of modality that affects data size, the software was structured to enable this flexibility.

The software was developed for the ESP32-S3 microcontroller used on the Unexpected Maker TinyS3 module. Firmware was implemented using Espressif's ESP-IDF framework, which provides a C/C++ development environment for configuring, building, and flashing applications directly onto the MCU. ESP-IDF also includes FreeRTOS support, allowing the embedded software to be structured as multiple concurrent tasks. FreeRTOS is a lightweight real-time operating system commonly used in edge devices for timing, resource management, and task scheduling.

The firmware was organized into two primary FreeRTOS tasks:

- **Sensor data acquisition:** Reads the ADC values from the connected sensor modules over the I^2C bus at a consistent rate.
- **Sensor data transmission:** Sends data to a host computer where it can be further

processed and stored.

The firmware was built using Espressif's pre-written I^{2}C and BLE drivers. These drivers provided the lower-level functionality needed to communicate with the ADCs over the I^{2}C bus and transmit data wirelessly over BLE. Using these drivers as a base allowed the firmware development to focus on the higher-level system structure, including task separation, timing, and data transfer between threads. The software project file layout is shown in Figure 7.

As shown in Figure 8, the firmware separates sensor acquisition and data transmission into independent tasks to prevent communication delays from interfering with sampling since data transfer over Bluetooth or USB can experience variable latency depending on connection stability, host-side processing, or buffer availability. By isolating the acquisition task from the transmission task, the system can prioritize consistent sensor sampling while allowing collected data to be sent asynchronously.

Data is passed between the acquisition and transmission tasks using a single-element FreeRTOS queue. The queue provides a task-safe communication mechanism by managing access to shared data internally, preventing race conditions that could occur if both tasks directly read from and write to the same memory location. The queue is stored in RAM and is typically allocated from heap memory when created dynamically using FreeRTOS functions such as xQueueCreate().

In this system, the acquisition task updates the queue using xQueueOverwrite(), allowing each new sensor sample to replace the previous sample when the queue is full. The transmission task then reads the most recent available sample when it is ready to send data. This design prioritizes real-time responsiveness over preserving every individual sample. Since USB or Bluetooth transmission may introduce variable delays, a longer queue could allow older, stale samples to accumulate and be transmitted after newer sensor readings are already available, increasing latency between the physical gesture and the transmitted data. By storing only the latest sample, the system ensures that downstream tasks operate on the most current FMG sensor state.

This structure allows the acquisition task to

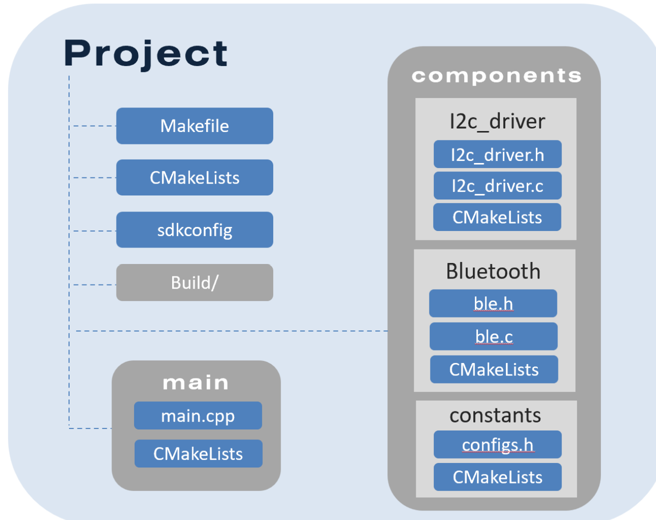

Figure 7. Diagram of software project structure.

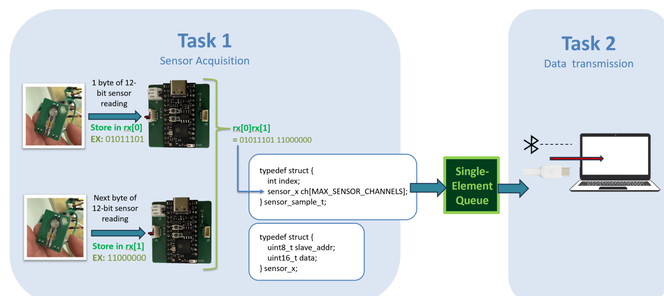

Figure 8. High-level data flow diagram illustrating the two-task firmware architecture. Task 1 (Sensor Acquisition) reads two bytes of each 12-bit ADC reading from each sensor over I^{2}C, assembling them into a `sensor_sample_t` struct that is written to a single-element FreeRTOS queue. Task 2 (Data Transmission) reads from the queue and forwards the data to a host computer via Bluetooth or USB.

continue sampling at a consistent rate without being blocked by the slower or more variable timing of the transmission task. As a result, the system can preserve real-time sensor collection behavior where stale data is less useful than the latest available measurement. This approach is also well suited for future expansion where the system response should reflect the user's current muscle activity rather than delayed or stale buffered samples, including applications that rely on on-board gesture classification and closed-loop gesture-based control.

IV. Validation

To verify hardware integration, each FMG sensor module was connected to the MCU over the shared

I^2C bus and the system was powered on. Upon initialization, the firmware enumerates all connected sensor modules and logs each detected device's slave address to the terminal. Figure 9 shows an example terminal output confirming that five sensor modules were successfully detected on the bus, each identified by a unique 7-bit I^2C slave address. No bus conflicts or address collisions were observed across the tested configurations, confirming that the ADC121C021 address configuration scheme described in Section III functions correctly in practice. Both USB and Bluetooth transmission modes were verified during this testing phase. Successful BLE connectivity was confirmed through Bluetooth discovery logs, which showed the device advertising and pairing correctly with the host computer. USB connectivity was verified through direct serial communication with the host.

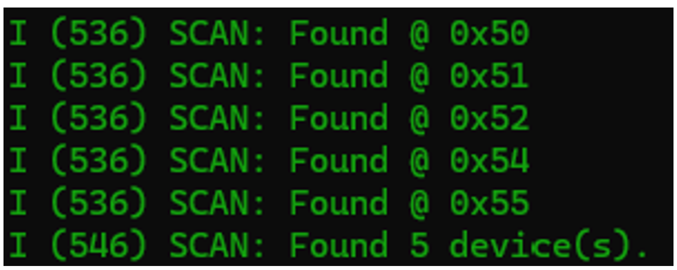

Figure 9. Terminal log of slave sensor module devices brought up on the I^2C bus line.

To validate the sensor hardware, a single sensor was manually compressed by applying direct finger pressure to simulate the range of force expected during muscle contraction. The maximum voltage output under finger compression measured at approximately 2.75 V, slightly below the 3.3 V reference, likely due to resistive losses in the sensor circuit. This range was sufficient to distinguish varying levels of muscle activity across gestures.

To validate the complete data pipeline, sensor data was collected from one participant wearing the device with five FMG sensor module daughterboards (Figure 10). The participant performed three cycles of a 3-second hand gesture followed by a 3-second rest period. A MATLAB script was developed to receive the incoming data stream over either USB or Bluetooth, reconstruct the 12-bit ADC values from the two bytes per sensor channel, and log and plot the resulting voltage signals across all active channels. Among the gestures tested, the fist contraction (Figure 11) produced the largest and most consistent voltage response. Figure 12 shows a voltage to time plot of the FSR sensor readings recorded across five channels during the hand open and close sequence. The results confirm that the system successfully captures and transmits sensor data in a form suitable for offline analysis and gesture classification, consistent with the design targets outlined in Section II.

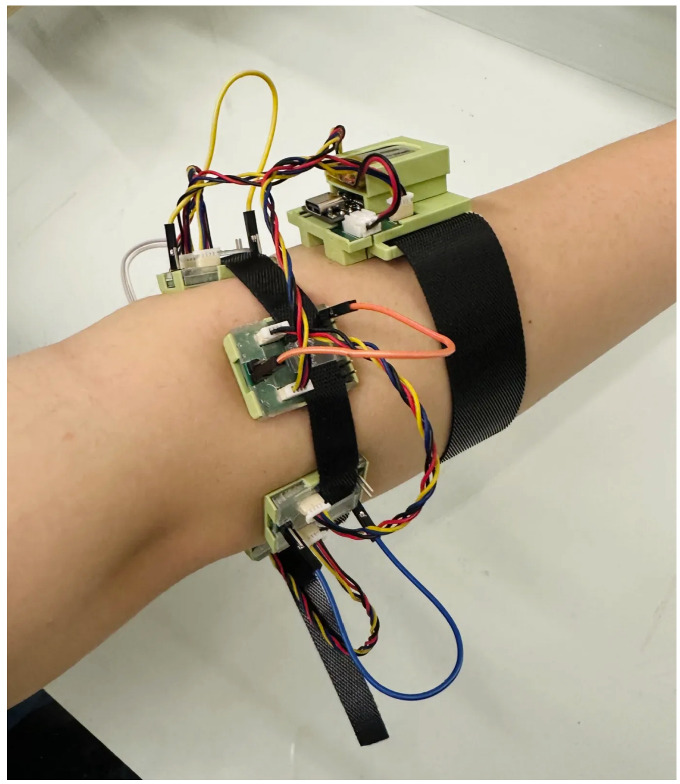

Figure 10. Completed assembly of band in mechanical casing.

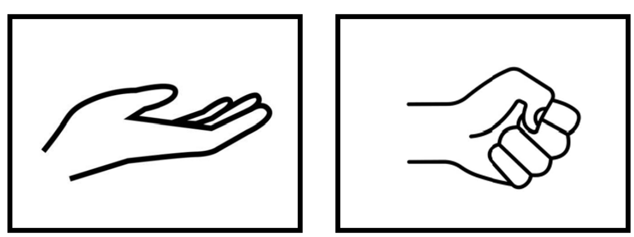

Figure 11. Gesture classes used in data collection. Participant performed a single repeated motion of opening and closing the hand, transitioning between a rest position (left, fingers fully extended and relaxed) and a fist contraction (right, fingers flexed into a closed fist).

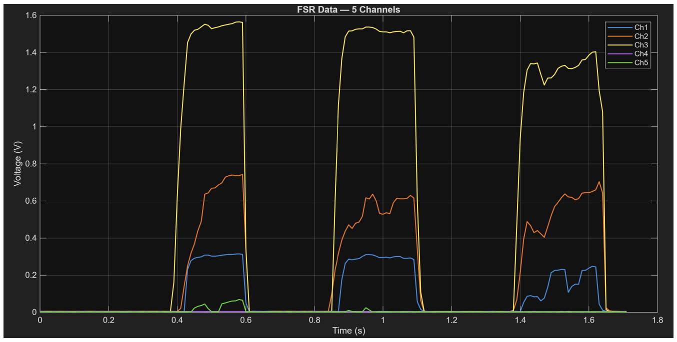

Figure 12. FSR voltage recordings from 5 sensors during repeated fist contractions. Participant performed 3 cycles of 3-second contraction followed by 3-second rest. Ch3 (yellow) shows the highest pressure response ($\sim$1.5V), consistent with placement over the primary flexor region.

TABLE I
DEVELOPMENT COST ESTIMATION

Item	**Qty.**	**Cost per Unit**	**Total Cost**
ESP32-TinyS3	1	$20.00	$20.00
PCB: MCU Daughterboard	1	$77.78	$77.78
PCB: FMG Sensor Module	10	$113.64	$113.64
FSR400	8	$2.99	$23.92
LiPo Battery	1	$9.99	$9.99
PDMS Domes	8	$5.99	$5.99
4-Wire JST-SH Cables	8	$1.70	$13.60
Velcro Strap	1	$6.20	$6.20
		Total	**$271.12**

V. Conclusion & Future Work

Based on our requirements for a successful project, we were successful in creating an FMG band that satisfies the design targets: Accessibility, Low Power Operation, Simple Circuitry, and Sensor Modularity. The total development cost came to $271.12 USD (Table I), which was below $300.00 and well within the range of an accessible, low-cost system compared to commercial sEMG alternatives. The device is powered entirely by a 3.7V LiPo battery and is able to sustain multiple data collection trials. Up to nine sensor modules can be configured and connected via a shared I^2C bus, with a sampling rate that falls within the range of 6 Hz to 1 kHz reported in existing FMG literature [5]. Placing the electronics within a mechanical encasing enabled us to test the device on an able-bodied participant. The fist-relax gesture produced clear and consistent voltage responses across all active sensor channels, demonstrating that the system successfully captures distinguishable muscle activity patterns. These results are promising and suggest the platform is ready to be validated across a broader participant pool and expanded gesture set. These qualities demonstrate that a capable, portable, and modular FMG data collection platform can be created at a fraction of the cost of existing commercial systems.

The band design establishes a foundation for future expansion, with additional opportunities for refinement that can be explored. Since the sensor modules are modular in the sense that they are easily removed or reattached, the system provides flexibility to integration sensing modalities that go beyond the physical forces exerted from FMG sensing. For example, future versions of the band could incorporate EMG sensor modules, enabling mixed FMG-EMG system within the same wearable platform.

Another potential expansion is enabling real-time control through on-board MCU classification. Rather than only transmitting sensor data to a host computer for storage or offline analysis, the MCU could perform feature extraction and classification directly on the device. This would allow the band to recognize different gestures in real time and use the classified output for downstream control applications, such as interacting with computer interfaces, robotic systems, or rehabilitation gamified applications.

Additional improvements could also be made to the mechanical packaging of the band. The current prototype uses a Velcro strap that wraps around the user's forearm. While this provides adjustability, it can also introduce variability in the amount of pressure applied to the sensors during use. This is partly due to the limited precision of Velcro adjustment. The PCB design could also be further optimized for the band's mechanical packaging. In the current design, some pressure points from the band may rest on parts of the PCB rather than directly over the pressure sensors. Future iterations could refine the PCB layout and enclosure so that applied force is better applied onto the active sensing region of the FSR. This could improve muscle pressure sensitivity, increase signal consistency, and make the band more reliable for repeated FMG measurements.

A more comprehensive improvement would be to design a custom MCU host PCB for the band. This would provide greater control over power consumption and space management of component layout, allowing footprint optimization of the motherboard for wearable use. A custom MCU host PCB would also enable more flexibility for future sensing unit expansion. For example, if EMG sensing or other body-sensing modalities are introduced, the MCU host PCB could be modified to include the required analog front-end circuitry, additional ADC channels, communication interfaces, or power management features.

Overall, this work sets up an open-source foundation for accessible force myography research and consumer-oriented control applications. By eliminating the need for expensive commercial acquisition hardware and enabling wireless data collection through Bluetooth or a simple USB connection, the system lowers the barrier to entry for both researchers and end-users. This ultimately brings gesture-based wearable control closer to everyday consumer use.

VI. Availability

The full implementation is available on GitHub: https://github.com/lwong370/fmg_grasp_classification.

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